



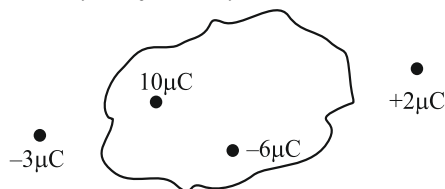
# Lakshya JEE 2027

KPP - 05

Electrostatics

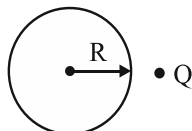
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1. In a particular system, if number of electric field lines associated by 1 C charge is  $10^\circ$ . Then net number of electric field lines passing through the given closed surface is  $n \times 10^3$  find  $n$ .

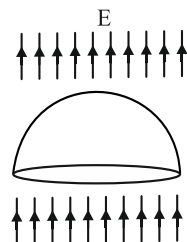


2. **Statement-1:** A charge is outside the Gaussian sphere of radius  $R$ . Then electric field on the surface of sphere is zero. and

**Statement-2:** As  $\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$ , for the sphere  $q_{in}$  is zero, so  $\oint \vec{E} \cdot d\vec{s} = 0$ .

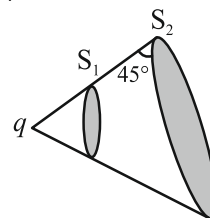


- (1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
(2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
(3) Statement-1 is true, statement-2 is false.  
(4) Statement-1 is false, statement-2 is true.
3. A hemispherical surface (half of a spherical surface) of radius  $R$  is located in a uniform electric field  $E$  that is parallel to the axis of the hemisphere. What is the magnitude of the electric flux through the hemisphere surface?

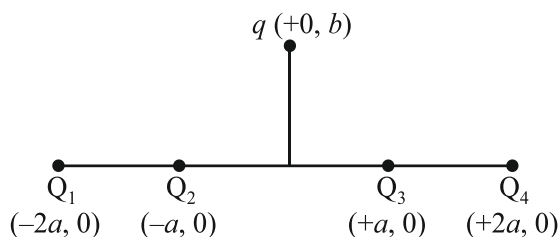


- (1) 0  
(2)  $4\pi R^2 E/3$   
(3)  $2\pi R^2 E$   
(4)  $\pi R^2 E$

4. In the given figure flux through surface  $S_1$  is  $\phi_1$  and through  $S_2$  is  $\phi_2$ . Which is correct?



- (1)  $\phi_1 = \phi_2$   
(2)  $\phi_1 > \phi_2$   
(3)  $\phi_1 < \phi_2$   
(4) None of these
5. Four charges  $Q_1, Q_2, Q_3$  and  $Q_4$  of same magnitude are fixed along the  $x = -2a, -a, +a$  and  $+2a$ , respectively. A positive charge  $q$  is placed on the positive  $y$ -axis at a distance  $b > 0$ . Four options of the signs of these charges are given in List-I. The direction of the forces on the charge  $q$  is given in List-II. Match List-I with List-II and select the correct answer using the code given below the lists.



| List-I |                                         | List-II |      |
|--------|-----------------------------------------|---------|------|
| P      | $Q_1, Q_2, Q_3, Q_4$ all positive       | 1       | $+x$ |
| Q      | $Q_1, Q_2$ positive $Q_3, Q_4$ negative | 2       | $-x$ |
| R      | $Q_1, Q_4$ positive $Q_2, Q_3$ negative | 3       | $+y$ |
| S      | $Q_1, Q_3$ positive $Q_2, Q_4$ negative | 4       | $-y$ |

- (1)  $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$   
(2)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$   
(3)  $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$   
(4)  $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

6. The electric field in a region is given by  $\vec{E} = \left( \frac{3}{5} E_0 \hat{i} + \frac{4}{5} E_0 \hat{j} \right) \frac{N}{C}$ . The ratio of flux of reported field through the rectangular surface of area  $0.2 \text{ m}^2$  (parallel to  $y-z$  plane) to that of the surface of area  $0.3 \text{ m}^2$  (parallel to  $x-z$  plane) is  $a : b$ , where  $a = \underline{\hspace{2cm}}$ . [Here  $\hat{i}, \hat{j}$  are  $\hat{k}$  unit vectors along  $x, y$  and  $z$ -axes respectively]

7. The electric field in a region is given  $\vec{E} = \frac{2}{5} E_0 \hat{i} + \frac{3}{5} E_0 \hat{j}$  with  $E_0 = 4.0 \times 10^3 \frac{N}{C}$ . The flux of this field through a rectangular surface area  $0.4 \text{ m}^2$  parallel to the  $Y-Z$  plane is  $\underline{\hspace{2cm}} \text{ Nm}^2 \text{ C}^{-1}$ .

8. A particle of mass  $1 \text{ mg}$  and charge  $q$  is lying at the mid-point of two stationary particles kept at a distance ' $2 \text{ m}$ ' when each is carrying same charge ' $q$ '. If the free charged particle is displaced from its equilibrium position through distance ' $x$ ' ( $x \ll 1 \text{ m}$ ). The particle executes SHM. Its angular frequency of oscillation will be  $\underline{\hspace{2cm}} \times 10^5 \text{ rad/s}$  if  $q^2 = 10 \text{ C}^2$ .

9. A uniformly charged disc of radius  $R$  having surface charge density  $\sigma$  is placed in the  $xy$  plane with its center at the origin. Find the electric field intensity along the  $z$ -axis at a distance  $Z$  from origin:

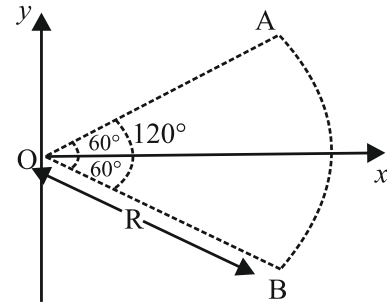
(1)  $E = \frac{\sigma}{2\epsilon_0} \left( 1 - \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$

(2)  $E = \frac{\sigma}{2\epsilon_0} \left( 1 + \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$

(3)  $E = \frac{2\epsilon_0}{\sigma} \left( \frac{Z}{(Z^2 + R^2)^{1/2}} + Z \right)$

(4)  $E = \frac{\sigma}{2\epsilon_0} \left( \frac{Z}{(Z^2 + R^2)} + \frac{1}{Z^2} \right)$

10. Figure shows a rod AB, which is bent in a  $120^\circ$  circular arc of radius  $R$ . A charge  $(-Q)$  is uniformly distributed over rod AB. What is the electric field  $\vec{E}$  at the centre of curvature O?

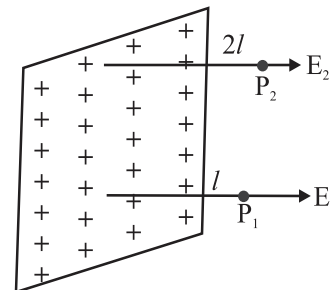


- (1)  $\frac{3\sqrt{3}Q}{8\pi\epsilon_0 R^2} (\hat{i})$  (2)  $\frac{3\sqrt{3}Q}{8\pi^2\epsilon_0 R^2} (\hat{i})$   
 (3)  $\frac{3\sqrt{3}Q}{16\pi^2\epsilon_0 R^2} (\hat{i})$  (4)  $\frac{3\sqrt{3}Q}{8\pi^2\epsilon_0 R^2} (-\hat{i})$

11. A charge of  $4 \mu\text{C}$  is to be divided into two parts. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be:

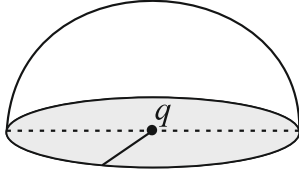
- (1)  $1 \mu\text{C}$  and  $3 \mu\text{C}$  (2)  $2 \mu\text{C}$  and  $2 \mu\text{C}$   
 (3)  $0 \mu\text{C}$  and  $4 \mu\text{C}$  (4)  $1.5 \mu\text{C}$  and  $2.5 \mu\text{C}$

12. In the figure, a very large plane sheet of positive charge is shown.  $P_1$  and  $P_2$  are two points at distance  $l$  and  $2l$  from the charge distribution. If  $\sigma$  is the surface charge density, then the magnitude of electric fields  $E_1$  and  $E_2$  at  $P_1$  and  $P_2$  respectively are:



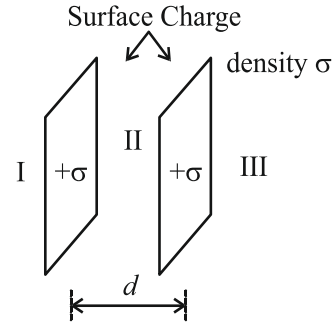
- (1)  $E_1 = \sigma/\epsilon_0, E_2 = \sigma/2\epsilon_0$   
 (2)  $E_1 = 2\sigma/\epsilon_0, E_2 = \sigma/\epsilon_0$   
 (3)  $E_1 = E_2 = \sigma/2\epsilon_0$   
 (4)  $E_1 = E_2 = \sigma/\epsilon_0$

13. If a charge  $q$  is placed at the centre of a closed hemispherical non-conducting surface, the total flux passing through the flat surface would be:



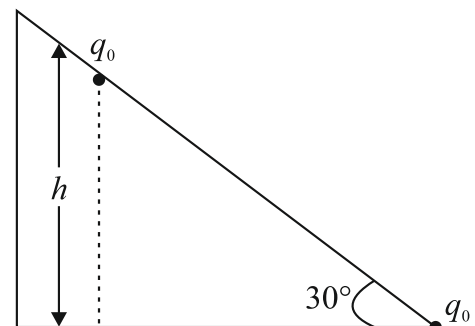
- (1)  $\frac{q}{\epsilon_0}$  (2)  $\frac{q}{2\epsilon_0}$   
 (3)  $\frac{q}{4\epsilon_0}$  (4)  $\frac{q}{2\pi\epsilon_0}$
14. Two point charges A and B of magnitude  $+8 \times 10^{-6} \text{ C}$  and  $-8 \times 10^{-6} \text{ C}$  respectively are placed at a distance  $d$  apart. The electric field at the middle point O between the charges is  $6.4 \times 10^4 \text{ NC}^{-1}$ . The distance ' $d$ ' between the point charges A and B is:  
 (1) 2.0 m (2) 3.0 m  
 (3) 1.0 m (4) 4.0 m
15. A positive charge particle of 100 mg is thrown in opposite direction to a uniform electric field of strength  $1 \times 10^5 \text{ NC}^{-1}$ . If the charge on the particle is  $40 \mu\text{C}$  and the initial velocity is  $200 \text{ ms}^{-1}$ , how much distance it will travel before coming to the rest momentarily:  
 (1) 1 m (2) 5 m  
 (3) 10 m (4) 0.5 m
16. A point charge of  $10 \mu\text{C}$  is placed at the origin. At what location on the X-axis should a point charge of  $40 \mu\text{C}$  be placed so that the net electric field is zero at  $x = 2 \text{ cm}$  on the X-axis?  
 (1)  $x = 6 \text{ cm}$  (2)  $x = 4 \text{ cm}$   
 (3)  $x = 8 \text{ cm}$  (4)  $x = -4 \text{ cm}$
17. A point charge  $q_1 = 4q_0$  is placed at origin. Another point charge  $q_2 = -q_0$  is placed at  $x = 12 \text{ cm}$ . Charge of proton is  $q_0$ . The proton is placed on x-axis so that the electrostatic force on the proton is zero. In this situation, the position of the proton from the origin is \_\_\_\_\_ cm.

18. Let  $\sigma$  be the uniform surface charge density of two infinite thin plane sheets shown in figure. Then the electric fields in three different region  $E_I$ ,  $E_{II}$  and  $E_{III}$  are:



- (1)  $\vec{E}_I = \frac{2\sigma}{\epsilon_0} \hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{2\sigma}{\epsilon_0} \hat{n}$   
 (2)  $\vec{E}_I = 0, \vec{E}_{II} = \frac{\sigma}{\epsilon_0} \hat{n}, \vec{E}_{III} = 0$   
 (3)  $\vec{E}_I = \frac{\sigma}{2\epsilon_0} \hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{2\epsilon_0} \hat{n}$   
 (4)  $\vec{E}_I = -\frac{\sigma}{\epsilon_0} \hat{n}, \vec{E}_{II} = 0, \vec{E}_{III} = \frac{\sigma}{\epsilon_0} \hat{n}$
19. As shown in the figure, a configuration of two equal point charges ( $q_0 = +2\mu\text{C}$ ) is placed on an inclined plane. Mass of each point charge is 20 g. Assume that there is no friction between charge and plane. For the system of two point charges to be in equilibrium (at rest) the height  $h = x \times 10^{-3} \text{ m}$ . The value of  $x$  is \_\_\_\_.

(Take  $\frac{1}{4\pi\epsilon_0}$  as  $9 \times 10^9 \text{ N m}^2\text{C}^{-2}$ ,  $g = 10 \text{ ms}^{-1}$ )



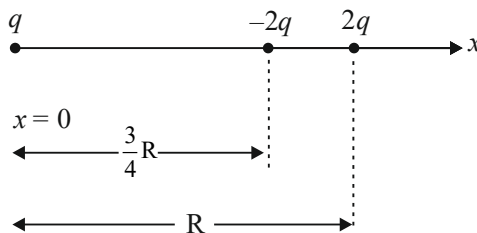
20. A  $10 \mu\text{C}$  charge is divided into two parts and placed at 1 cm distance so that the repulsive force between them is maximum. The charges of the two parts are:

(1)  $7 \mu\text{C}, 3 \mu\text{C}$   
 (2)  $8 \mu\text{C}, 2 \mu\text{C}$   
 (3)  $5 \mu\text{C}, 5 \mu\text{C}$   
 (4)  $9 \mu\text{C}, 1 \mu\text{C}$

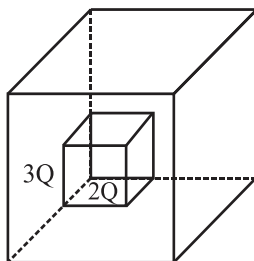
21. Two small spheres each of mass 10 mg are suspended from a point by threads 0.5 m long. They are equally charged and repel each other to a distance of 0.20 m.

The charge on each of the sphere is  $\frac{a}{21} \times 10^{-8} \text{C}$ . The value of 'a' will be \_\_\_\_\_. [Given  $g = 10 \text{ ms}^{-2}$ ].

22. Three point charges  $q, -2q$  and  $2q$  are placed on x-axis at a distance  $x = 0, x = \frac{3}{4}R$  and  $x = R$  respectively from origin as shown. If  $q = 2 \times 10^{-6} \text{C}$  and  $R = 2 \text{ cm}$ , the magnitude of net force experienced by the charge  $-2q$  is \_\_\_\_\_ N.



23.  $C_1$  and  $C_2$  are two hollow concentric cubes enclosing charges  $2Q$  and  $3Q$  respectively as shown in figure. The ratio of electric flux passing through  $C_1$  and  $C_2$  is:

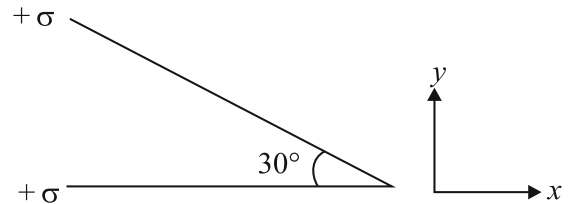


(1) 2 : 5                      (2) 5 : 2  
 (3) 2 : 3                      (4) 3 : 2

24. Two charges  $q$  and  $3q$  are separated by a distance ' $r$ ' in air. At a distance  $x$  from charge  $q$ , the resultant electric field is zero. The value of  $x$  is:

(1)  $\frac{(1+\sqrt{3})}{r}$                       (2)  $\frac{r}{3(1+\sqrt{3})}$   
 (3)  $\frac{r}{(1+\sqrt{3})}$                       (4)  $r(1+\sqrt{3})$

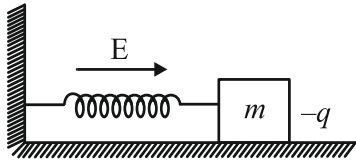
25. Two infinite planes each with uniform surface charge density  $+\sigma$  are kept in such a way that the angle between them is  $30^\circ$ . The electric field in the region shown between them is given by:



(1)  $\frac{\sigma}{\epsilon_0} \left[ \left( 1 + \frac{\sqrt{3}}{2} \right) \hat{y} + \frac{\hat{x}}{2} \right]$   
 (2)  $\frac{\sigma}{2\epsilon_0} \left[ \left( 1 - \frac{\sqrt{3}}{2} \right) \hat{y} - \frac{\hat{x}}{2} \right]$   
 (3)  $\frac{\sigma}{2\epsilon_0} \left[ (1 + \sqrt{3}) \hat{y} + \frac{\hat{x}}{2} \right]$   
 (4)  $\frac{\sigma}{2\epsilon_0} \left[ (1 + \sqrt{3}) \hat{y} - \frac{\hat{x}}{2} \right]$

26. A thin infinite sheet charge and an infinite line charge of respective charge densities  $+\sigma$  and  $+\lambda$  are placed parallel at 5 m distance from each other. Points 'P' and 'Q' are at  $\frac{3}{\pi} \text{ m}$  and  $\frac{4}{\pi} \text{ m}$  perpendicular distance from line charge towards sheet charge, respectively. ' $E_P$ ' and ' $E_Q$ ' are the magnitudes of resultant electric field intensities at point 'P' and 'Q', respectively. If  $\frac{E_P}{E_Q} = \frac{4}{a}$  for  $2|\sigma| = |\lambda|$ . Then the value of  $a$  is \_\_\_\_\_.

27. A block of mass  $m$  containing a net negative charge  $-q$  is placed on a frictionless horizontal table and is connected to a wall through an unstretched spring of spring constant  $k$  as shown. If horizontal electric field  $E$  parallel to the spring is switched on, then the maximum compression of the spring is  $\frac{nqE}{k}$ . Find the value of  $n$ .



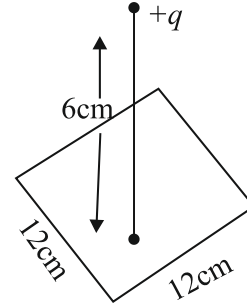
28. A flat, square surface with sides of length  $L$  is described by the equations  $x = L, 0 \leq y \leq L, 0 \leq z \leq L$ . The electric flux through the square due to a positive point charge  $q$  located at the origin ( $x = 0, y = 0, z = 0$ ) is  $\frac{q}{x \epsilon_0}$ . What is the value of  $x$ ?

29. A solid sphere of radius  $R$  has a charge  $Q$  distributed in its volume with a charge density  $\rho = kr^a$  where  $k$  and  $a$  are constants and  $r$  is the distance from its centre. If the electric field at  $r = R/2$  is  $1/8$  times that at  $r = R$  find the value of  $a$ .

30. Two electrons each are fixed at a distance ' $2d$ '. A third charge proton placed at the midpoint is displaced slightly by a distance  $x$  ( $x \ll d$ ) perpendicular to the line joining the two fixed charges. Proton will execute simple harmonic motion having angular frequency: ( $m$  = mass of charged particle).

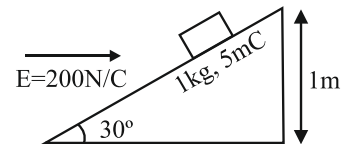
(1)  $\left(\frac{2q^2}{\pi \epsilon_0 md^3}\right)^{\frac{1}{2}}$  (2)  $\left(\frac{\pi \epsilon_0 md^3}{2q^2}\right)^{\frac{1}{2}}$   
 (3)  $\left(\frac{q^2}{2\pi \epsilon_0 md^3}\right)^{\frac{1}{2}}$  (4)  $\left(\frac{2\pi \epsilon_0 md^3}{q^2}\right)^{\frac{1}{2}}$

31. A point charge of  $+12 \mu\text{C}$  is at a distance  $6 \text{ cm}$  vertically above the centre of side of a square of side  $12 \text{ cm}$  as shown in figure. The magnitude of the electric flux through the square will be  $\_\_\_\_ \times 10^3 \text{ Nm}^2/\text{C}$ .



32. An inclined plane making an angle of  $30^\circ$  with the horizontal is placed in a uniform horizontal electric field  $200 \frac{\text{N}}{\text{C}}$  as shown in the figure. A body of mass  $1 \text{ kg}$  and charge  $5 \text{ mC}$  is allowed to slide down from rest at a height of  $1 \text{ m}$ . If the coefficient of friction is  $0.2$ , find the time taken by the body to reach the bottom.

$[g = 9.8 \text{ m/s}^2, \sin 30^\circ = \frac{1}{2}; \cos 30^\circ = \frac{\sqrt{3}}{2}]$

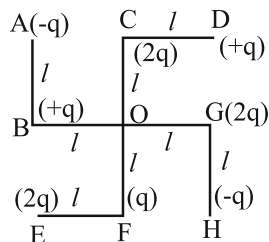


- (1)  $0.92 \text{ s}$  (2)  $0.46 \text{ s}$   
 (3)  $2.3 \text{ s}$  (4)  $1.3 \text{ s}$

33. Two identical tennis balls each having mass ' $m$ ' and charge ' $q$ ' are suspended from a fixed point by threads of length ' $l$ '. What is the equilibrium separation when each thread makes a small angle ' $\theta$ ' with the vertical?

(1)  $x = \left(\frac{q^2 l}{2\pi \epsilon_0 mg}\right)^{\frac{1}{2}}$  (2)  $x = \left(\frac{q^2 l}{2\pi \epsilon_0 mg}\right)^{\frac{1}{3}}$   
 (3)  $x = \left(\frac{q^2 l^2}{2\pi \epsilon_0 m^2 g}\right)^{\frac{1}{3}}$  (4)  $x = \left(\frac{q^2 l^2}{2\pi \epsilon_0 m^2 g^2}\right)^{\frac{1}{3}}$

34. What will be the magnitude of electric field at point O as shown in figure? Each side of the figure is  $l$  and perpendicular to each other?

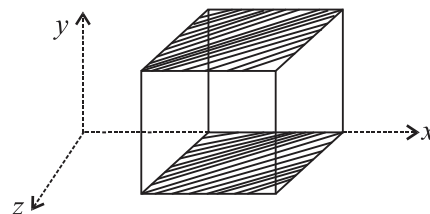


- (1)  $\frac{1}{4\pi\epsilon_0} \frac{q}{l^2}$
- (2)  $\frac{1}{4\pi\epsilon_0} \frac{q}{(2l^2)} (2\sqrt{2} - 1)$
- (3)  $\frac{q}{4\pi\epsilon_0 (2l^2)}$
- (4)  $\frac{2q}{4\pi\epsilon_0 2l^2} (\sqrt{2})$
35. Choose the incorrect statement:
- (A) The electric lines of force entering into a Gaussian surface provide negative flux.
- (B) A charge ' $q$ ' is placed at the centre of a cube. The flux through all the faces will be the same.
- (C) In a uniform electric field net flux through a closed Gaussian surface containing no net charge, is zero.
- (D) When electric field is parallel to a Gaussian surface, it provides a finite non-zero flux.

Choose the most appropriate answer from the options given below.

- (1) (C) and (D) only
- (2) (B) and (D) only
- (3) (D) only
- (4) (A) and (C) only

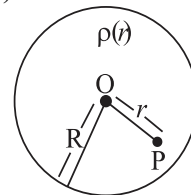
36. A cube is placed inside an electric field,  $\vec{E} = 150y^2 \hat{j}$ . The side of the cube is 0.5 m and is placed in the field as shown in the given figure. The charge inside the cube is:



- (1)  $3.8 \times 10^{-11} \text{ C}$  (2)  $8.3 \times 10^{-11} \text{ C}$
- (3)  $3.8 \times 10^{-12} \text{ C}$  (4)  $8.3 \times 10^{-12} \text{ C}$
37. The volume charge density of a sphere of radius 6 m is  $2 \mu\text{C cm}^{-3}$ . The number of lines of force per unit surface area coming out from the surface of the sphere is  $\times 10^{10} \text{ NC}^{-1}$ . (Given: Permittivity of vacuum  $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$ ).
38. Three point charges of magnitude  $5\mu\text{C}$ ,  $0.16\mu\text{C}$  and  $0.3\mu\text{C}$  are located at the vertices A, B, C of a right angled triangle whose sides are  $AB = 3 \text{ cm}$ ,  $BC = 3\sqrt{2} \text{ cm}$  and  $CA = 3 \text{ cm}$  and point A is the right angle corner. Charge at point A experiences  $\times 10^4 \text{ N}$  of electrostatic force due to the other two charges.
39. A spherically symmetric charge distribution is considered with charge density varying as:

$$\rho(r) = \begin{cases} \rho_0 \left( \frac{3}{4} - \frac{r}{R} \right) & \text{for } r \leq R \\ \text{zero} & \text{for } r > R \end{cases}$$

Where,  $r(r < R)$  is the distance from the centre O (as show in figure). The electric field at point P will be:



- (1)  $\frac{\rho_0 r}{4\epsilon_0} \left( \frac{3}{4} - \frac{r}{R} \right)$  (2)  $\frac{\rho_0 r}{3\epsilon_0} \left( \frac{3}{4} - \frac{r}{R} \right)$
- (3)  $\frac{\rho_0 r}{4\epsilon_0} \left( 1 - \frac{r}{R} \right)$  (4)  $\frac{\rho_0 r}{5\epsilon_0} \left( 1 - \frac{r}{R} \right)$

40. A vertical electric field of magnitude  $4.9 \times 10^5 \text{ N/C}$  just prevents a water droplet of a mass  $0.1 \text{ g}$  from falling. The value of charge on the droplet will be:

(Given  $g = 9.8 \text{ m/s}^2$ ).

- (1)  $1.6 \times 10^{-9} \text{ C}$   
 (2)  $2.0 \times 10^{-9} \text{ C}$   
 (3)  $3.2 \times 10^{-9} \text{ C}$   
 (4)  $0.5 \times 10^{-9} \text{ C}$

41. Two identical charged particles each having a mass  $10 \text{ g}$  and charge  $2.0 \times 10^{-7} \text{ C}$  placed on a horizontal table with a separation of  $L$  between them such that they stay in limited equilibrium. If the coefficient of friction between each particle and the table is  $0.25$ , find the value of  $L$ . [Use  $g = 10 \text{ ms}^{-2}$ ].

- (1)  $12 \text{ cm}$  (2)  $10 \text{ cm}$   
 (3)  $8 \text{ cm}$  (4)  $5 \text{ cm}$

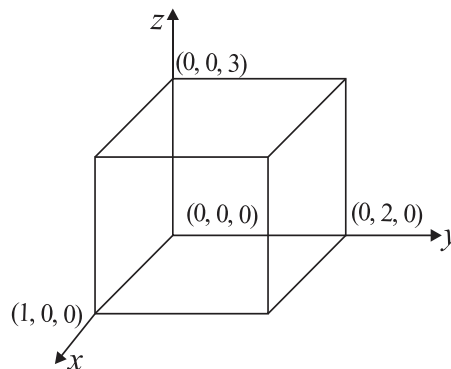
42. If two charges  $q_1$  and  $q_2$  are separated with distance ' $d$ ' and placed in a medium of dielectric constant  $K$ . What will be the equivalent distance between charges in air for the same electrostatic force?

- (1)  $d\sqrt{K}$  (2)  $K\sqrt{d}$   
 (3)  $1.5d\sqrt{K}$  (4)  $2d\sqrt{K}$

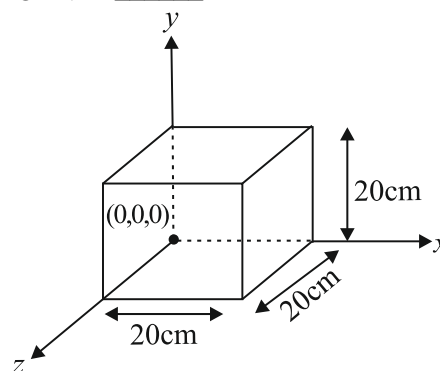
- 43 (a). Force between two point charges  $q_1$ , and  $q_2$  placed in vacuum at ' $r$ ' cm apart is  $F$ . Force between them when placed in a medium having dielectric  $K = 5$  at ' $r/5$ ' cm apart will be:

- (1)  $F/25$  (2)  $5F$   
 (3)  $F/5$  (4)  $25F$

- 43 (b). As shown in figure, a cuboid lies in a region with electric field  $\vec{E} = 2x^2\hat{i} - 4y\hat{j} + 6\hat{k}$ . The magnitude of charge within the cuboid is  $n\epsilon_0 \text{ C N/C}$ . The value of  $n$  is \_\_\_\_\_ (if dimension of cuboid is  $1 \times 2 \times 3 \text{ m}^3$ )



44. Expression for an electric field is given by  $\vec{E} = 4000x^2\hat{i} \frac{\text{V}}{\text{m}}$ . The electric flux through the cube of side  $20 \text{ cm}$  when placed in electric field (as shown in the figure) is \_\_\_\_\_  $\text{V cm}$ .



45. An electric field is given by  $(6\hat{i} + 5\hat{j} + 3\hat{k}) \text{ N/C}$ . The electric flux through a surface area  $30\hat{i} \text{ m}^2$  lying in  $YZ$ -plane (in SI unit) is:

- (1)  $90$  (2)  $150$   
 (3)  $180$  (4)  $60$



## Answer Key

- |            |               |
|------------|---------------|
| 1. (4)     | 24. (3)       |
| 2. (4)     | 25. (2)       |
| 3. (4)     | 26. (6)       |
| 4. (1)     | 27. (2)       |
| 5. (1)     | 28. (24)      |
| 6. (1)     | 29. (2)       |
| 7. (640)   | 30. (3)       |
| 8. (6000)  | 31. (226)     |
| 9. (1)     | 32. (4)       |
| 10. (2)    | 33. (2)       |
| 11. (2)    | 34. (2)       |
| 12. (3)    | 35. (3)       |
| 13. (2)    | 36. (2)       |
| 14. (2)    | 37. (45)      |
| 15. (4)    | 38. (17)      |
| 16. (1)    | 39. (3)       |
| 17. (24)   | 40. (2)       |
| 18. (4)    | 41. (1)       |
| 19. (300)  | 42. (1)       |
| 20. (3)    | 43. (a.) (2)  |
| 21. (20)   | 43. (b.) (12) |
| 22. (5440) | 44. (640)     |
| 23. (1)    | 45. (3)       |



PW Web/App - <https://smart.link/7wwosivoicgd4>

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